🏛️ The Foundation: The Probability Space

You're spot on that there are two ways to think about this. Your first definition is the formal, mathematical one.

Your notes: `Defined on the 3 tuple ( theta -> sample space - set of outcomes , sigma -> set of possible events , p ) , also called sample space`

My Detailed Explanation:

This 3-tuple is the bedrock of modern probability. It's formally called a **Probability Space**. Let's refine the terms slightly:

1. **Sample Space (Ω):** You called this θ (theta). This is correct! It's the set of **all possible individual outcomes** of an experiment.
   - **Example:** Flipping a coin. $\Omega = \{\text{Heads}, \text{Tails}\}$.
   - **Example:** Rolling a six-sided die. $\Omega = \{1, 2, 3, 4, 5, 6\}$.
2. **Event Space ($\mathcal{F}$ or σ):** You called this σ (sigma). This is the set of **all events** we care about. An **event** is a *set of one or more outcomes* (a subset of Ω).
   - **Example (Die Roll):**
     - The outcome "rolling a 2" is represented by the event $\{2\}$.
     - The event "rolling an even number" is the set $\{2, 4, 6\}$.
     - The event "rolling a number greater than 3" is the set $\{4, 5, 6\}$.
   - The event space $\mathcal{F}$ contains *all* these sets, including the empty set $\emptyset$ (the "impossible event") and the entire sample space Ω (the "certain event").
3. **Probability Measure (P):** You called this p. This is a **function** that assigns a real-numbered probability (between 0 and 1) to every event in the event space $\mathcal{F}$.
   - $P : \mathcal{F} \to [0, 1]$
   - **Example (Fair Die Roll):**
     - $P(\{2\}) = 1/6$
     - $P(\{2, 4, 6\}) = 3/6 = 1/2$
     - $P(\emptyset) = 0$
     - $P(\Omega) = P(\{1, 2, 3, 4, 5, 6\}) = 1$

📜 The Axiomatic Definition

Your notes: `Axiomatic : a probability function p is a function that maps each element in theta ( events ) to a real value`

My Detailed Explanation:

You've got the right idea, but let's connect it to the 3-tuple. The "axiomatic" definition refers to the three rules that the probability measure P *must* obey. These are called the **Kolmogorov Axioms**.

A function P is a valid probability measure if and only if it satisfies:

1. **Axiom 1: Non-negativity**
   For any event A (which is in $\mathcal{F}$), its probability must be zero or positive.
   $$P(A) \geq 0$$
2. **Axiom 2: Normalization**
   The probability of the entire sample space Ω (the "certain event") is 1.
   $$P(\Omega) = 1$$
3. **Axiom 3: Countable Additivity**
   If you have a collection of **disjoint** events $A_1, A_2, A_3, \ldots$ (meaning no two events can happen at the same time, e.g., "rolling a 1" and "rolling a 2"), the probability that *at least one* of them occurs is the sum of their individual probabilities.
   $$P(A_1 \cup A_2 \cup A_3 \cup \ldots) = P(A_1) + P(A_2) + P(A_3) + \ldots$$

This third axiom is the most powerful. It's what allows us to calculate the probability of complex events (like "rolling an even number") by breaking them down into simple, disjoint outcomes ($P(\{2\}) + P(\{4\}) + P(\{6\})$).

🎲 The Random Variable (RV)

Your notes:

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Random variable -> function that maps events in sample space to real values , Probability of random variable getting a particular value is sum o
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My Detailed Explanation:

This is a very common point of confusion. Let's clarify the mapping.

A **Random Variable (RV)**, usually written X, is a function that maps **outcomes in the sample space (Ω)** to real numbers ($\mathbb{R}$).

- **Formal Mapping:** $X : \Omega \to \mathbb{R}$

It's a "variable" because it takes on different numerical values. It's "random" because the outcome from Ω that it maps is uncertain.

Example: Experiment = Flip 2 Coins

- Sample Space $\Omega = \{HH, HT, TH, TT\}$
- Let's define a Random Variable X = "the number of heads".
- This RV X is a function that does the following mapping:
  - $X(HH) = 2$
  - $X(HT) = 1$
  - $X(TH) = 1$
  - $X(TT) = 0$

Now, let's look at your second point, which is **exactly correct**.

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Probability of random variable getting a particular value is sum of prob of events resulting in that value of R.V
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- What is $P(X = 1)$?
- We look at all outcomes in Ω that X maps to the value 1. This is the event $\{HT, TH\}$.
- Therefore, $P(X = 1) = P(\{HT, TH\})$.
- Using Axiom 3 (Additivity), since HT and TH are disjoint:
  $P(X = 1) = P(HT) + P(TH)$
- If the coin is fair, $P(HT) = 1/4$ and $P(TH) = 1/4$.
- So, $P(X = 1) = 1/4 + 1/4 = 1/2$.

📊 Discrete vs. Continuous RVs

Discrete RV

(Takes on a countable number of values, like 0, 1, 2, 3...)

- **Your notes:** `Distribution fn fx is just P( X = x)`
  - **Clarification:** This is called the **Probability Mass Function (PMF)**. It's often written as $p_X(x)$. It's a function that gives the probability of *each specific value*.
  - **Example (2 coin flips):** $p_X(0) = 1/4$, $p_X(1) = 1/2$, $p_X(2) = 1/4$.
- **Your notes:** `Cdf is P ( X <= x)`
  - **Correct.** This is the **Cumulative Distribution Function (CDF)**, written $F_X(x)$. It "accumulates" probability.
  - **Example (2 coin flips):**
    - $F_X(0) = P(X \leq 0) = 1/4$
    - $F_X(1) = P(X \leq 1) = P(X = 0) + P(X = 1) = 1/4 + 1/2 = 3/4$
    - $F_X(1.5) = P(X \leq 1.5) = P(X = 0) + P(X = 1) = 3/4$
    - $F_X(2) = P(X \leq 2) = P(X = 0) + P(X = 1) + P(X = 2) = 1$
  - The CDF for a discrete RV is a **step function**. It jumps up at each possible value.

Continuous RV

(Takes on an uncountable number of values in a range, like height or temperature)

- **Your notes:** `P( X = x) is just 0`
  - **Exactly.** The probability of a continuous RV landing on *one specific, exact value* (e.g., $X = 3.1415926...$) is zero. There are infinitely many possible values, so any single one has zero probability.
  - This is why we only talk about probabilities for *ranges* (e.g., $P(3 \leq X \leq 4)$).
- **Your notes:** `Cdf is F(x) = P ( X<= x) and its differentiation gives us distribution function`
  - **Perfect.** The CDF, $F_X(x)$, is defined the same way.

  - The function you get by differentiating it is called the **Probability Density Function (PDF)**, written $f_X(x)$.
  - $f_X(x) = \frac{d}{dx}F_X(x)$
- **Your notes:** `Which unlike in the case of discrete doesnt give us prob but gives likelihood`
  - **This is a critical insight!** A PDF value $f_X(x)$ is **not a probability**. It can even be greater than 1.
  - It represents **probability density** or **likelihood**. A higher $f_X(x)$ means values *around* x are more likely.
  - To get an actual probability, you must **integrate** the PDF over a range:

$$P(a \le X \le b) = \int_a^b f_X(x)\,dx$$

  - This integral is the "area under the curve" of the PDF, which *is* the probability.
- **Your notes:** `Cdf for crv is right continuous`
  - **Correct.** This is a property of *all* CDFs (both discrete and continuous).

📈 Expectation and Variance

These are properties that *summarize* a random variable's distribution.

Expectation (E[X])

The **Expectation** (or *mean*) is the long-term average value of the RV. It's the "center of mass" of its distribution.

- **For Discrete RVs:** It's a weighted sum.

$$E[X] = \sum_x x \cdot P(X = x)$$

- **For Continuous RVs:** (As you noted) It's a weighted integral.

$$E[X] = \int_{-\infty}^{\infty} x \cdot f_X(x)\,dx$$

Expectation of a Function, g(X)

Your notes on this are excellent. This is often called the **Law of the Unconscious Statistician (LOTUS)**.
If $Y = g(X)$ (e.g., $Y = X^2$), how to find $E[Y]$?

- **Way 1 (The "Unconscious" or Easy Way):** (Your first method)
  - **Discrete:** $E[g(X)] = \sum_x g(x) \cdot P(X = x)$
  - **Continuous:** $E[g(X)] = \int_{-\infty}^{\infty} g(x) \cdot f_X(x)\,dx$
    You just "plug in" $g(x)$ for x in the expectation formula. You *don't* need to find the distribution of $g(X)$.
- **Way 2 (The "Conscious" or Hard Way):** (Your second method)
  i. Let $Y = g(X)$.
  ii. Find the full probability distribution (PMF or PDF) of Y, which is $f_Y(y)$. This can be very difficult.
  iii. Apply the basic definition of expectation to Y: $E[Y] = \int y \cdot f_Y(y)\,dy$.
    Both ways give the *exact same answer*. Way 1 is almost always easier.

Variance (Var(X))

Variance measures the "spread" of the distribution. It's the *average squared distance from the mean*.

- **Your notes:** `Variance E[ X - E[x] ]`
  - **Small correction:** You're missing the *square*. The definition is the **expected squared deviation**.
  - Let $\mu = E[X]$ (the mean).
  - **Definition:** $\mathrm{Var}(X) = E[(X - \mu)^2]$
  - If we use LOTUS with $g(X) = (X - \mu)^2$, this becomes:
    - **Discrete:** $\sum (x - \mu)^2 \cdot P(X = x)$
    - **Continuous:** $\int (x - \mu)^2 \cdot f_X(x)\,dx$
- **Added Stuff (Useful Computational Formula):**
  It's almost always easier to calculate variance using this formula, which is derived from the definition:

$$\mathrm{Var}(X) = E[X^2] - (E[X])^2$$

  In words: "The expectation of the square, minus the square of the expectation."

This is a fantastic start. You've covered all the core concepts.

Ready for the next set of notes when you are!